



miR-29c-3p promotes senescence of human mesenchymal stem cells by targeting CNOT6 through p53–p21 and p16–pRB pathways



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ABSTRACT

Mesenchymal stem cells (MSCs) are important seed cells for tissue engineering and are promising targets for cell-based therapies. However, the replicative senescence of MSCs during *in vitro* culture limits their research and clinical applications. The molecular mechanisms underlying the replicative senescence of MSCs are not fully understood. Evidence suggests that miRNAs play important roles in replicative senescence. A microarray analysis found that the miR-29c-3p level was significantly increased during the MSC senescence process. In our study, we investigated the roles of miR-29c-3p in senescence of MSCs. We cultured MSCs for long periods of time, up and down-regulated the miR-29c-3p expression in MSCs, and examined the senescent phenotype changes. The over-expression of miR-29c-3p led to enhanced senescence-associated-β-galactosidase (SA-β-gal) staining, senescence associated secretory phenotype (SASP), senescence associated heterochromatic foci (SAHF), reduced proliferation ability, retarded osteogenic differentiation and corresponding changes in senescence markers, whereas the miR-29c-3p down-regulation had the opposite results. Dual-luciferase reporter assays demonstrated that CNOT6 is the target gene of miR-29c-3p. Knockdown of CNOT6 confirmed its inhibitory effects on the senescence of MSCs. In addition, Western blot results showed that both the p53–p21 and the p16–pRB pathways were activated during the miR-29c-3p-induced senescence of MSCs. In conclusion, our results demonstrate that miR-29c-3p promotes the senescence of MSCs by targeting CNOT6 through p53–p21 and p16–pRB pathways and highlight the contribution of post-transcriptional regulation to stem cell senescence.

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1. Introduction

Human mesenchymal stem cells (hMSCs) are a group of adult pluripotent stem cells that are capable of self-renewing and differentiating into both mesodermal and non-mesodermal cell lineages, such as osteocytes, chondrocytes, adipocytes, myocytes and hepatocytes [1–4]. Because of their appreciable advantages, MSCs are considered as an important source for tissue engineering and are quite promising for cell-based therapeutic strategies. However, long periods of *in vitro* culture inevitably lead to replicative senescence in hMSCs. Replicative senescence is the process in which cells are permanently deprived of the capacity for division while maintaining viability [5,6]. In the senescent state, MSCs show impaired proliferation and differentiation capacity

[7,8], and thus senescence could hinder to the application of MSCs in clinical trials.

Until now, the molecular mechanisms underlying cellular senescence remain poorly understood. Several genome-wide analyses provided important hints about the mechanisms underlying senescence. miRNAs, which function at the post-transcriptional level, are a group of small and non-coding RNAs that are 18–25 nucleotides in length and generally repress target gene expression by degrading mRNA or preventing translation [9]. miRNAs have been reported to play roles in many complex cellular and biological processes, including proliferation, differentiation, apoptosis and cancer [10]. miRNAs have also been implicated in cellular senescence [11–13]. Furthermore, Wagner and his colleagues performed a microarray analysis and found five significantly increased miRNAs (hsa-miR-371, hsa-miR-369-3p, hsa-miR-29c, hsa-miR-499 and hsa-miR-217) during the replicative senescence of MSCs [14]. The up-regulation of miR-29c was validated by qPCR. Several studies have identified miR-29c in particular as a modulator of cellular senescence [15,16]. However, the specific roles of miR-29c in the senescence of hMSCs are largely unknown.

In our study, we investigated the effects of miR-29c-3p in the senescence of hMSCs and identified CNOT6 as its target gene. Additionally,

Abbreviations: hMSCs, human mesenchymal stem cells; SA-β-gal, senescence-associated-β-galactosidase; ANOVA, analysis of variance; CCK-8, Cell Counting Kit-8; WB, Western blot; IF, immunofluorescence.

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both p53–p21 and p16–pRB pathways were activated during miR-29c-3p induced senescence of hMSCs.

2. Materials and methods

2.1. Culture of hMSCs

Human bone marrow-derived mesenchymal stem cells were purchased from Cyagen Biosciences Incorporation (HUXMA-01001, Oricell™, Cyagen). MSC complete culture medium (HUXMA-90011, Oricell™, Cyagen) was used to cultivate hMSCs; this medium consists of 440 mL Human Mesenchymal Stem Cell Basal Medium, 50 mL Human Mesenchymal Stem Cell-Qualified Fetal Bovine Serum, 5 mL penicillin–streptomycin and 5 mL glutamine. For the long duration culture, hMSCs were seeded at a density of 2×10^5 cells/75 cm², and were subcultured when the cells reached 80% confluence.

2.2. miRNA and siRNA cell transfection assays

miR-29c-3p was over-expressed by transfecting hMSCs with Agomir-29c-3p (B06002, GenePharma) and the corresponding negative control (NC) (Agomir-NC, B04008, GenePharma). Compared with miRNA mimics, Agomirs are able to up-regulate miRNAs for a longer period of time because of their chemically modified structure and enhanced stability [17]. In addition, the mirVana™ miR-29c-3p inhibitor (mirVana-29c-3p, 4464084 mirVana™, Invitrogen) and the corresponding NC (mirVana-NC, 4464076 mirVana™, Invitrogen) were used to down-regulate the expression of miR-29c-3p in hMSCs. CNOT6 was inhibited in hMSCs using commercially available siRNAs and their corresponding NCs (si-CNOT6 and si-NC) (Invitrogen). The transfection reagent Lipofectamine2000 (11668027, Invitrogen) was used to perform Agomir, mirVana and siRNA transfections according to the manufacturer's instructions. Briefly, hMSCs were seeded at a density of 5×10^4 cells/well. After hMSCs reached approximately 50% confluence, Agomirs, mirVanas and siRNAs were added into the wells. The final incubation concentrations of Agomirs, mirVanas and siRNAs were all 100 nM. After a 6-h incubation, the culture medium in the wells was replaced by complete culture medium. Then, the hMSCs were cultured for another 48 and 96 h before being used for transcriptional and translational analyses, respectively.

2.3. Real-time quantitative PCR (qPCR)

Reverse transcriptase (RT)-qPCR analysis was performed to investigate the differential mRNA and miRNA expression levels of the genes of interest. GAPDH was chosen to be an internal control. Total RNA was extracted and used to generate cDNA using a Takara kit (RR047A, Japan) according to the manufacturer's instructions. The quality of the total RNA was examined by a spectrophotometer (Nanodrop 2000, Thermo Scientific) at 260 nm and 280 nm. Primers were designed using Primer

Premier 6.0 software. The cycling parameters of the RT reaction were 37 °C for 15 min and 85 °C for 5 s. Next, the cDNA was subjected to qPCR with SYBR® Green (Applied Biosystems, USA) staining. The StepOnePlus Real-Time PCR System (Applied Biosystems) was used to perform and analyze the qPCR assays. The primers used in the qPCR assays are listed in Table 1.

2.4. Western blot (WB) analysis

A Western blot was performed as previously described [18]. Briefly, a lysis buffer was used to lyse hMSCs under different experimental preconditions. The lysis buffer contained 50 mM Tris (pH 7.6), 150 mM NaCl, 1% Triton X-100, 1% deoxycholate, 0.1% SDS, 1 mM PMSF, 0.2% Aprotinin and freshly added protease/phosphatase inhibitor cocktail. After the protein concentration was determined using an Enhanced BCA Protein Assay Kit (P0010S, Beyotime), the equal-protein samples were separated by 7.5–15% sodium dodecylsulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to polyvinylidene fluoride (PVDF, No. ISEQ00010, Immobilon) membranes at 200 mA for 1 h. Ten percent non-fat dried milk was used to block the membrane in TBST for 2 h at room temperature. After blocking, the membrane was incubated with primary antibodies at 4 °C overnight. The primary antibodies used to assess proteins of interest were as follows: p16 (ab81278, Abcam, 1:1000); p21 (#2947, Cell Signaling Technology, 1:1000); PTGS2 (#12282, Cell Signaling Technology, 1:1000); GAPDH (#5174, Cell Signaling Technology, 1:8000); CNOT6 (#13415, Cell Signaling Technology, 1:1000); p53 (sc-126, Santa Cruz, 1:1000); pRB (sc-50, Santa Cruz, 1:700) and phosphorylated pRB (ppRB, sc-377528, Santa Cruz, 1:700). GAPDH was used as the internal control. The protein blots were visualized using a chemiluminescence detection kit (#34095, Thermo Scientific, USA).

2.5. Proliferation assay

The cell proliferation capacity was detected using Cell Counting Kit-8 (GC614, Dojindo Laboratories) according to the manufacturer's instructions. Briefly, the cells were seeded into 96-well culture plates at a density of 1500 cells/well. After 24-h incubation, the culture medium in each well was replaced by 100 µL fresh medium containing 10 µL CCK-8. Then, the plate was incubated for 2 h at 37 °C. The optical density (OD) values were measured in at least triplicates at 450 nm using a microplate reader (Thermo Scientific, USA). The CCK-8 assay was repeated every 24 h and a growth curve was obtained to compare the proliferative capacity among samples.

2.6. Immunofluorescence (IF) assay

hMSCs were seeded on glass chamber slides, fixed with 4% paraformaldehyde for 25 min and permeabilized with 0.1% Triton (Amresco, Sigma Aldrich, USA). Then hMSCs were blocked with 10% normal goat

Table 1
Primers used in qPCR assays.

Gene symbol	Forward primers (5'–3')	Reverse primers (5'–3')
GAPDH	TGGGTGTGAACCATGAGAAGT	TGAGTCCTCCACGATACCAA
U6	CTCGCTTCGGCAGCACAA	AACGCTTCACGAATTTGCGT
miR-29c-3p	ACACTCCAGCTGGGTAGCACATTGAAAT	TGGTGTGCTGGAGTCG
p16	TTCCCCACTACCGTAAATGT	GCTCACTCCAGAAAACCTCAAC
p21	CTCAAGGCCCGCTCTACAT	AATGCCAGCACTCTTAGGAA
PTGS2	TTCTCTCTGTGCCTGATGATT	AAACTGATGCGTGAAGTGCTG
AKAP9	ACAACAAGCAAGAAGAGAAAAGGA	GCTGCTGTTGATGAGTGAGG
SCIN	AAAGGCGGTCTGAAATACAAGG	GCCAAGGTCAATGATGAAGCA
EDN1	CCAGAGAGCGTTATGTGACCC	TGTTTTGAACGAGGACGCTG
CXCL12	CAGAGCCAACGTCAAGCATC	CTTCGGGTCAATGCACACTT
SOD2	GGTTTTGGGGTATCTGGGCTC	CAGGTTGTTACAGTAGGCCG
CCND1	GACTCCAAATCTCAATGAAGCCA	GGTAAGCGTGAGCCGTGTTT

serum (Solarbio, China) at room temperature for 25 min and incubated with primary antibodies against p16 (1:100), p21 (1:100), COX2 (1:100) and Alexa-Fluor-647- or TRITC-conjugated secondary antibodies (Invitrogen, USA). The cell nuclei were stained with DAPI (Beyotime, China) for 15 min to visualize senescence associated heterochromatic foci (SAHF). IF images were acquired under a confocal microscope (S75, Leica) and quantified using the Image J (Version 1.48u) software.

2.7. Senescence associated secretory phenotype (SASP) analysis

Subconfluent cultures after different treatments were washed and incubated in serum-free Dulbecco's modified Eagle medium (DMEM) for 24 h to prepare conditioned medium (CM), which was gathered and cell counted. ELISA assays were performed to detect several secreted proteins in CM according to the manufacturer's instructions, such as IL-1 α , IL-6, IL-8, MMP-3, GM-CSF, MCP-1 (HushangBio, Shanghai, China).

2.8. miRNA target gene prediction

Three target prediction algorithms, TargetScan 6.2 (<http://www.targetscan.org/>), PicTar (<http://pictar.mdc-berlin.de/>), and miRBase 21 (<http://www.mirbase.org/>), were used to search for the potential target genes of miR-29c-3p.

2.9. Dual-luciferase reporter assays

A dual-luciferase reporter vector (pmiR-RB-REPORT™, RiboBio Co., Ltd) was used to build the luciferase constructs. A fragment of the 3' untranslated region (3'UTR) of human CNOT6, which was complementary to miR-29c-3p, was amplified and cloned by PCR using two primers (F: 5'-GGCGGCTCGAGTCTGAACATAGGGGAGTGAAGTA-3'; R: 5'-AATGCGCGCCGTAGGCAGGCACAGTGGAGAT-3'). The PCR products were then inserted into the pmiR-RB-REPORT vector with the usage of two restriction enzymes (XhoI and NotI) to generate the wild-type luciferase construct (CNOT6-WT). To build the mutant construct, the mutant sequence of the miR-29c-3p binding site was cloned by PCR using two primers (F: 5'-GAAAGCCACCACGATGCAACAGACAAATTCTGA-3'; R: 5'-TCTGTTGCATCGTGGTGGCTTTCATATATC-3'). Then, the mutant luciferase construct (CNOT6-Mut) was generated using the PCR products in a similar fashion. The 293 T cells were seeded in 96-well plates at a density of 1.5×10^4 cells/well 24 h before transfection. CNOT6-WT or CNOT6-Mut were co-transfected with 50 nM miR-29c-3p mimics (miR10000681, RiboBio Co. Ltd.) or the NC (miR01201, RiboBio Co. Ltd.) into 293 T cells using Lipofectamine 2000 according to the manufacturer's instructions. The transfection was performed in triplicate. After a 48-h transfection, the hRluc and hluc luciferase activities were measured using a luminometer in the Dual-Glo® Luciferase Assay System (Promega, USA).

2.10. Senescence-associated- β -galactosidase (SA- β -gal) staining

SA- β -gal staining was performed using the Senescence β -Galactosidase Staining Kit (No. 9860S, Cell Signaling Technology) according to the manufacturer's instructions. Briefly, hMSCs were seeded at a density of 1.5×10^5 cells/well in 6-well plates. After the corresponding treatments, hMSC were washed twice with PBS and fixed with Fixative Solution for 15 min at room temperature. After fixation, the hMSCs were stained with 1 mL complete β -gal staining solution per well overnight at 37 °C. The complete β -gal staining solution contained Staining Solution, Solution A, Solution B and X-gal solution (final concentration of 1 mg/ml). After staining, the cell images were captured using a microscope (IX73, Olympus).

2.11. Osteogenic differentiation assay

The hMSCs were seeded at 1×10^5 cells/cm² in 6-well culture plates pre-coated with gelatin. The complete osteogenic differentiation medium (Cat. No. HUXMA-90021, Cyagen) consists of 175 mL basal medium, 20 mL fetal bovine serum, 2 mL penicillin-streptomycin, 2 mL glutamine, 400 μ L ascorbate, 2 mL β -glycerophosphate, and 20 μ L dexamethasone. After CESC reached 50%–70% confluence, the culture medium was replaced with 2 mL complete medium. The complete medium was replaced every 3 days and the cells were cultured for 14 days. After differentiation, the cells were fixed with 4% paraformaldehyde for 30 min at room temperature and stained with alizarin red working solution for 5 min. Images were captured under a light microscope (IX73, Olympus).

2.12. EdU incorporation assay

Cellular senescence was further validated by the co-staining of EdU and SA- β -gal. Cell proliferation was assayed using Cell-Light EdU DNA Cell Proliferation Kit (RiboBio, Guangzhou, China) according to the manufacturer's protocol. Briefly, 50 μ M EdU was added into the 24-well cultures and cells were incubated for 2 h. Next, the cells were fixed with 4% formaldehyde in PBS for 30 min. Then Apollo Staining reagents were added to react with EdU and Hoechst33342 was used to visualize the nuclei. The images were obtained using a fluorescence microscope (IX73, Olympus).

2.13. Statistical analysis

All of the experiments were performed at least in triplicate, and all experimental results were presented as the mean value $\pm$ standard deviation. The data were analyzed using a two-tailed Student's t-test and an analysis of variance (ANOVA). A value of $P < 0.05$ was treated as significant.

3. Results

3.1. Changes in senescent phenotype and miR-29c-3p in hMSCs after a long period of culture

First, we investigated the effects of replicative senescence on hMSC phenotype and function. To induce replicative senescence, we subcultured hMSCs repeatedly for at least eight weeks. The morphology of different generations of hMSCs (P2, P6 and P12) was compared, as shown in Fig. 1A. The early-passage hMSCs had a vigorous and healthy spindle shape, whereas the late-passage hMSCs lost normal viability and had an irregular shape. In addition, because an abundance of the β -gal enzyme accumulates in the lysosomes of senescent cells, SA- β -gal staining was performed on different generations of hMSCs, and the results showed that SA- β -gal activity increased gradually (Fig. 1B). To investigate the SASP of senescent hMSCs, we collected the CM of different generations of hMSCs and compared their secretome. The ELISA results showed that all detected proteins but IL-1 α remained low at first and increased gradually as the replicative senescence proceeded (Fig. 1C). The SAHF is also a senescence marker. In sequentially passaged hMSCs, we found that the percentage of SAHF-positive cells was elevated gradually (Fig. 1D). To evaluate the proliferative capacity of hMSCs, a Cell Counting Kit-8 (CCK-8) assay was performed on different generations of hMSCs. Fig. 1E indicates that the proliferative ability decreased as hMSCs became senescent, and late-passage hMSCs (P12) almost didn't divide at all. Several genes are believed to be closely associated with senescence, including p16, p21, PTGS2, SOD2, CXCL12, AKAP9, CCND1, EDN1, and SCIN [12]. qPCR results showed that p16, p21 and PTGS2, which are senescence markers, were significantly up-regulated in senescent hMSCs. Moreover, the SOD2 and CXCL12 levels decreased significantly in senescent hMSCs, and their deficiency has been

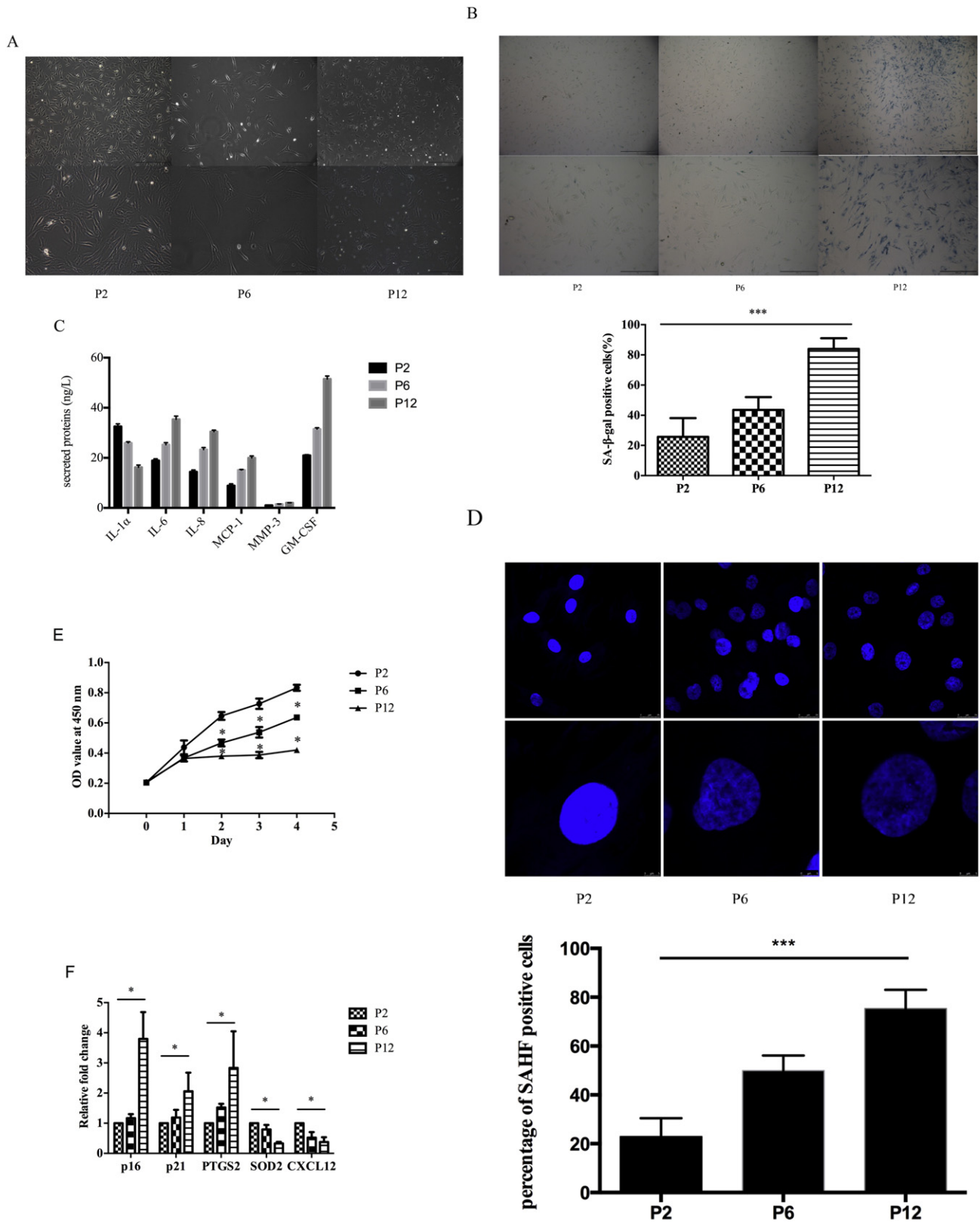


Fig. 1. Senescence phenotype of hMSCs after long periods of in vitro culture. hMSCs were subcultured for long periods to induce replicative senescence. (A) The morphologies of sequentially passed hMSCs (P2, P6 and P12) under a light microscope were shown. The lower panel is the high-magnification cell images. (B) The senescence-associated-β-galactosidase (SA-β-gal) staining of sequentially passed hMSCs (P2, P6 and P12) was shown. The rates of SA-β-gal-positive cells were calculated by counting the positive cells. The data were expressed as the mean ± standard deviation (S.D.). (C) The senescence-associated secretory phenotype (SASP) was detected by ELISA assays in sequentially passed hMSCs. (D) The senescence-associated heterochromatic foci (SAHF) were visualized using DAPI staining in sequentially passed hMSCs. The percentages of SAHF-positive hMSCs were calculated by counting the positive cells. The data were expressed as the mean ± S.D. (E) The proliferative capacity of sequentially passed hMSCs (P2, P6 and P12) was assessed by CCK-8 assays. The OD values were measured in triplicate at least 24 h. A growth curve was created to compare the proliferative capacities among the samples. (F) and (G) The expression levels of senescence-related genes were evaluated by qPCR and Western blot (WB) assays. Three samples (P2, P6 and P12) were compared. (H) The expression of miR-29c-3p in sequentially passed hMSCs (P2, P6 and P12) was examined by qPCR assays. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

G

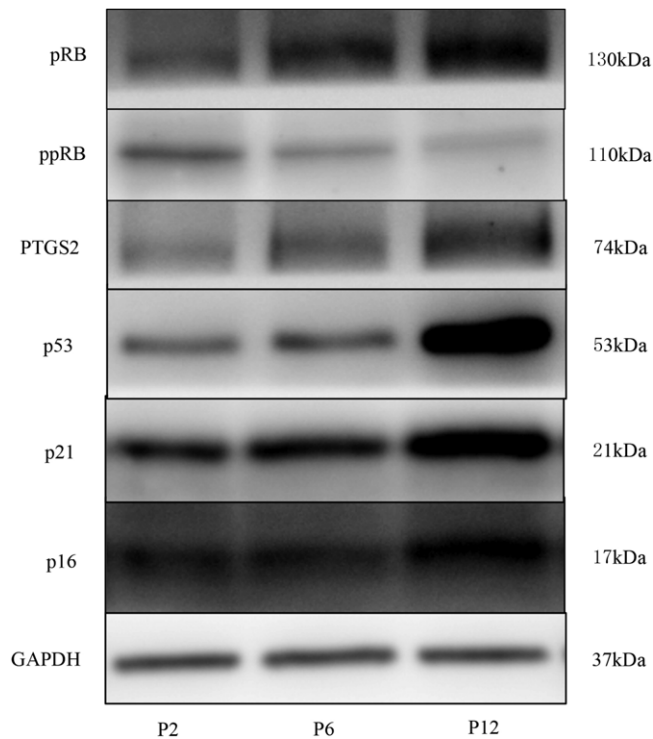
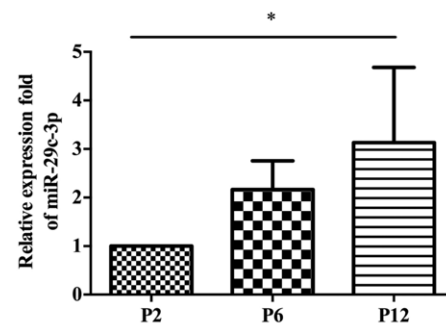


Fig. 1 (continued).

H



associated with cellular senescence [19,20] (Fig. 1F). Western blot (WB) results also suggested that the expression of p53, p21, p16, PTGS2 and pRB were all increased gradually, otherwise the expression of ppRB was decreased (Fig. 1G).

A previous microarray study reported that miR-29c-3p is up-regulated in senescent hMSCs [14]. To investigate the roles of miR-29c-3p in the replicative senescence of hMSCs, we assessed the expression level of miR-29c-3p in different generations of hMSCs by qPCR. The results showed that miR-29c-3p expression gradually increased as the passage number increased (Fig. 1H). The over-expression of miR-29c-3p suggested potential roles for this miRNA in senescent hMSCs.

3.2. Effects of miR-29c-3p up-regulation on the senescence of hMSCs

To evaluate the potential correlation between miR-29c-3p and the replicative senescence of hMSCs, we used chemically modified Agomirs (Agomir-29c-3p) to transfect P3 hMSCs to up-regulated the miR-29c-3p expression level. The qPCR results obtained after a 6-h transfection confirmed that the miR-29c-3p level was strongly increased in miR-29c-3p-transfected hMSCs (Fig. 2A). To determine the effects of miR-29c-3p, we first performed the co-staining of EdU and SA- β -gal on transfected hMSCs. The results demonstrated that the over-expression of miR-29c-3p significantly enhanced the SA- β -gal staining compared with that of the negative control (NC) group. Besides, EdU was not incorporated into the SA- β -gal positive hMSCs, indicating the senescent hMSCs were indeed not proliferating (Fig. 2B). To determine SASP of transfected hMSCs, ELISA assays were performed and results showed that the expression levels of secreted protein except IL-1 α were enhanced after Agomir-29c-3p transfection (Fig. 2C). In addition, more SAHF-positive cells are found in Agomir-29c-3p transfected hMSCs (Fig. 2D). To evaluate the proliferative ability, the CCK-8 assay was performed and repeated 7 times during a continuous 6-day period. The results showed that the proliferative capacity of hMSCs was inhibited after the Agomir-29c-3p transfection (Fig. 2E). To further confirm the effects of miR-29c-3p on the cellular senescence, the expression levels of

several senescence markers were evaluated using qPCR and IF assays. The qPCR results indicated that p16, p21, PTGS2, AKAP9, CCND1 and EDN1 were significantly increased after miR-29c-3p transfection (Fig. 2F). In addition, WB results showed that expression of p53, p21, p16, PTGS2 and pRB were all enhanced after Agomir-29c-3p transfection, except ppRB (Fig. 2G). In IF assays, the rates of p16-, p21- and PTGS2-positive cells after miR-29c-3p transfection were significantly higher than those of NC cells (Fig. 2H). Additionally, a previous study reported that MSCs in the senescent state had decreased differentiation potential, and miRNAs were involved in this process [21]. To determine the osteogenic potential of hMSCs transfected with Agomir-29c-3p, alizarin red staining and qPCR assays were performed. The results showed that the alizarin red staining was strongly inhibited after miR-29c-3p transfection compared with the staining in the NC group (Fig. 2I). The levels of three osteogenic markers, COL1A1, ALP and OCN were also significantly decreased after miR-29c-3p transfection (Fig. 2J). These results showed that miR-29c-3p promoted the replicative senescence and inhibited the osteogenic differentiation of hMSCs.

3.3. Effects of miR-29c-3p down-regulation on the senescent state of hMSCs

To investigate the effects of miR-29c-3p inhibition on the replicative senescence of hMSCs, we used the mirVana miRNA inhibitor (mirVana-29c-3p) to transfect P7 hMSCs to retard the endogenous expression of miR-29c-3p. After a 6-h transfection, we performed qPCR assays and found that the expression of miR-29c-3p in the mirVana-29c-3p group was significantly reduced compared with that in the NC group (Fig. 3A). The result validated the inhibitory effect of mirVana-29c-3p transfection on miR-29c-3p expression in hMSCs. Therefore, we performed co-staining of EdU and SA- β -gal to assess the effects of miR-29c-3p down-regulation. The SA- β -gal activity was significantly retarded in the mirVana-29c-3p group compared with that in the NC group. The co-staining also confirmed the growth arrest state of SA- β -gal positive hMSCs (Fig. 3B). Moreover, those secreted proteins were decreased after mirVana-29c-3p transfection, except IL-1 α (Fig. 3C). Similarly,

the rate of SAHF-positive hMSCs was also reduced in mirVana-29c-3p transfected hMSCs (Fig. 3D). Additionally, in CCK-8 assays, mirVana-29c-3p transfection significantly rescued the proliferative capacity (Fig. 3E). In terms of genes reported to be associated with senescence, the qPCR results showed that the p16, p21, PTGS2, AKAP9, SCIN and CCND1 levels decreased significantly after mirVana-29c-3p transfection, whereas the SOD2 and CXCL12 expression increased significantly (Fig. 3F). WB results showed that the expression of p53, p21, p16, PTGS2

and pRB was reduced after miR-29c-3p inhibition, while ppRB was increased (Fig. 3G). In IF assays, the rates of p16-, p21- and PTGS2-positive cells in the mirVana-29c-3p group were significantly lower than those in the NC group (Fig. 3H). Moreover, we performed alizarin red staining and qPCR assays to examine the effects of miR-29c-3p inhibition on the osteogenic capacity of hMSCs. The results indicated that after mirVana-29c-3p transfection, the alizarin red staining was enhanced, and the levels of osteogenic markers (COL1A1, ALP and OCN)

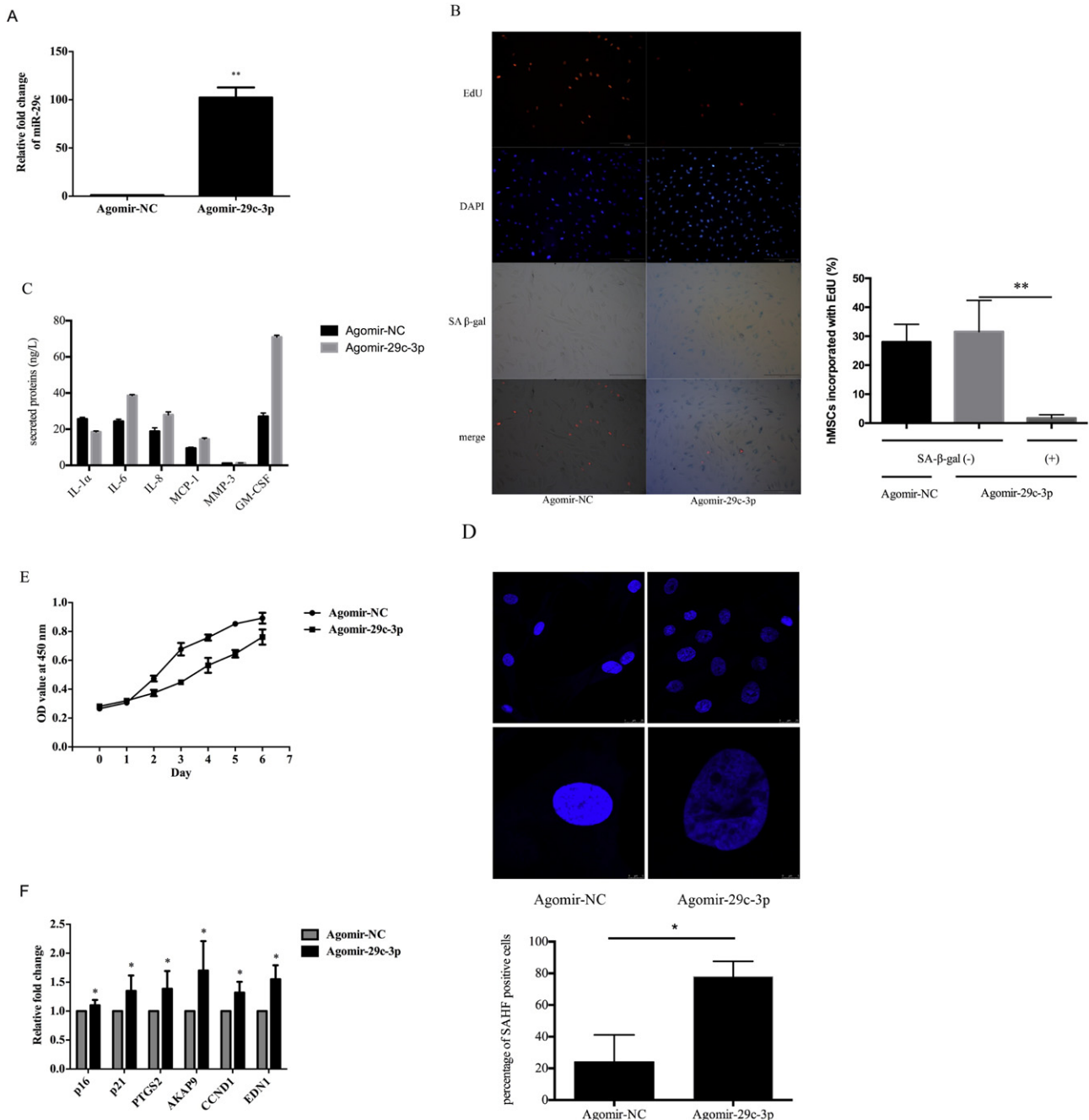


Fig. 2. Effects of miR-29c-3p over-expression on the senescence of hMSCs. Agomir-29c-3p was transfected into hMSCs to up-regulate the expression of miR-29c-3p. Agomir-NC was used as the negative control. (A) After transfection, the expression levels of miR-29c-3p in Agomir-NC and Agomir-29c-3p samples were compared by qPCR assays. (B) Co-staining of EdU and SA-βgal was performed in Agomir-NC and Agomir-29c-3p groups. The histogram showed the percentages of EdU-positive hMSCs between the two samples. (C) The SASP was detected by ELISA assays between Agomir-NC and Agomir-29c-3p groups. (D) The SAHF were visualized using DAPI staining between Agomir-NC and Agomir-29c-3p groups. The percentages of SAHF-positive hMSCs were calculated by counting the positive cells. The data were expressed as the mean $\pm$ S.D. (E) Changes in proliferative capacity due to miR-29c-3p over-expression were detected by CCK-8 assays. The OD values were measured in triplicate at least every 24 h. A growth curve was obtained to compare the proliferative capacities between samples. The expression levels of senescence-related genes after miR-29c-3p up-regulation were compared by (F) qPCR, (G) Western blot and (H) immunofluorescence (IF) assays. (I) The osteogenic differentiation of Agomir-NC and Agomir-29c-3p samples was compared by alizarin red staining. (J) Expression changes in osteogenic markers after miR-29c-3p up-regulation were assessed by qPCR assays. Data were expressed as the mean $\pm$ standard deviation (S.D.). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

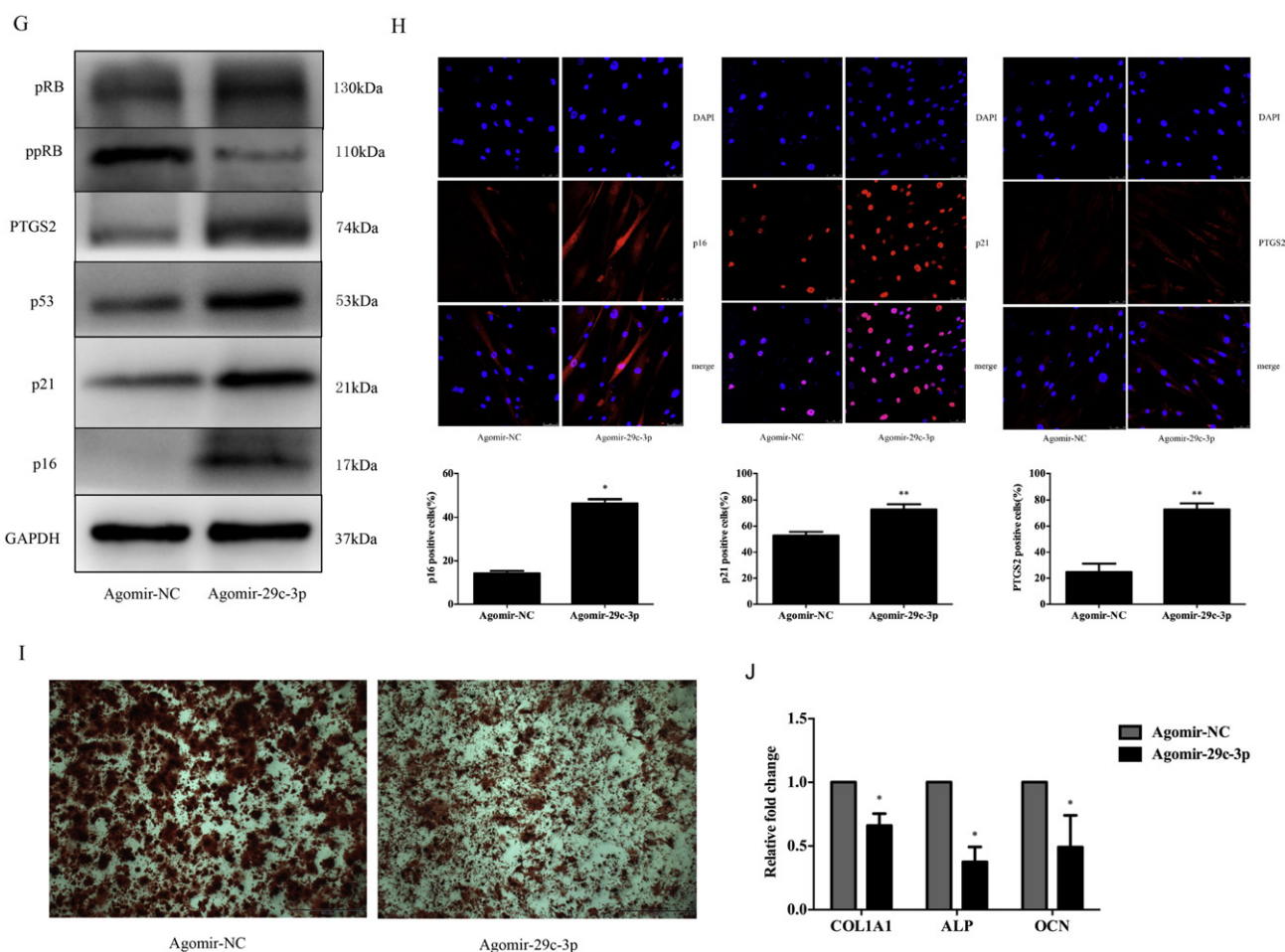


Fig. 2 (continued).

increased significantly (Fig. 3I and J). The results confirmed that down-regulation of miR-29c-3p partially rescued the osteogenic differentiation capacity of senescent hMSCs.

3.4. CNOT6 identified as the target gene of miR-29c-3p

To find genes that are targeted by miR-29c-3p and play a role in hMSC senescence, we used the three target prediction algorithm websites mentioned above to search the 3'UTR of putative genes for miR-29c-3p binding sites. Bioinformatics analysis indicated that CNOT6 was a candidate target gene of miR-29c-3p (Fig. 4A). Therefore, we performed dual-luciferase reporter assays to confirm that miR-29c-3p targets the CNOT6 3'UTR. The luciferase activity was significantly inhibited when CNOT6-WT was co-transfected with miR-29c-3p mimics compared with that after the NC co-transfection, whereas the inhibitory effect was abolished when the CNOT6 3'UTR was mutated (Fig. 4B). To further confirm that the CNOT6 3'UTR was the target site, we transfected hMSCs with Agomir-29c-3p or mirVana-29c-3p and examined CNOT6 expression using qPCR and WB assays. The results showed that the CNOT6 mRNA level was decreased in the Agomir-29c-3p group, but increased in the mirVana-29c-3p group (Fig. 4C). We thus inferred that the miR-29c-3p-induced inhibition of CNOT6 occurred at the transcription level. The WB results were consistent with the qPCR results (Fig. 4D).

3.5. Roles of CNOT6 in the senescence of hMSCs

To corroborate the functions of CNOT6 in the replicative senescence of hMSCs, we first investigated the change in CNOT6 expression during

hMSC replicative senescence using qPCR and WB assays. The results showed that as the replicative senescence proceeded, the expression level of CNOT6 decreased gradually (Fig. 5A and B), which suggested an inhibitory effect of CNOT6 on replicative senescence. Guided by these results, we therefore suppressed endogenous CNOT6 expression by transfecting hMSCs with small interfering (si) RNA (si-CNOT6). The qPCR results demonstrated that si-CNOT6 transfection effectively down-regulated the endogenous expression of CNOT6 (Fig. 5C). In SA- β -gal staining assays, si-CNOT6 transfection yielded a significant increase in SA- β -gal activity relative to that in the si-NC group (Fig. 5D). In terms of SASP, ELISA results showed elevated secreted proteins levels after si-CNOT6 transfection, except IL-1 α (Fig. 5E). What's more, the percentage of SAHF-positive hMSCs was also increased due to CNOT6 knockdown (Fig. 5F). In CCK-8 assays, the results suggested that si-CNOT6 transfection suppressed the proliferative capacity of hMSCs compared with that in the si-NC group (Fig. 5G). Similarly, we also examined several senescence markers after si-CNOT6 or si-NC transfection. The results showed that the mRNA levels of p16, p21, PTGS2, AKAP9, SCIN and CCND1 were significantly up-regulated in the si-CNOT6 transfection group (Fig. 5H). In addition, WB assays suggested that expression levels of p53, p21, p16, PTGS2 and pRB except ppRB were all increased after CNOT6 knockdown (Fig. 5I). We also performed immunofluorescence staining and found that the rates of p16- and p21-positive cells in the si-CNOT6 group were significantly higher than those in the si-NC group (Fig. 5J). Additionally, we performed qPCR assay to investigate CNOT6 changes during osteogenic differentiation. The results showed that expression of CNOT6 was increased during osteogenic differentiation, but the difference was not significant (Fig. 5K).

4. Discussion

hMSCs are practical and promising resources for tissue repair and regeneration in the tissue-engineering field. However, the concept of MSCs is a common misleading point. The current protocols for MSC isolation don't result in a homogeneous MSC population [22]. The MSC isolation methods are generally based on the criteria proposed by the

International Society for Cellular Therapy (ISCT) [23]. The criteria require that plastic adherence of MSCs under standard culture conditions; CD73, CD90 and CD105 positive; CD34 and CD45 negative; differentiation ability into osteocytes, chondrocytes and adipocytes. But isolation methods based on these criteria always lead to a heterogeneous cell population, which contain mesenchymal stem cells with diversified multipotential properties, committed progenitors, differentiated cells

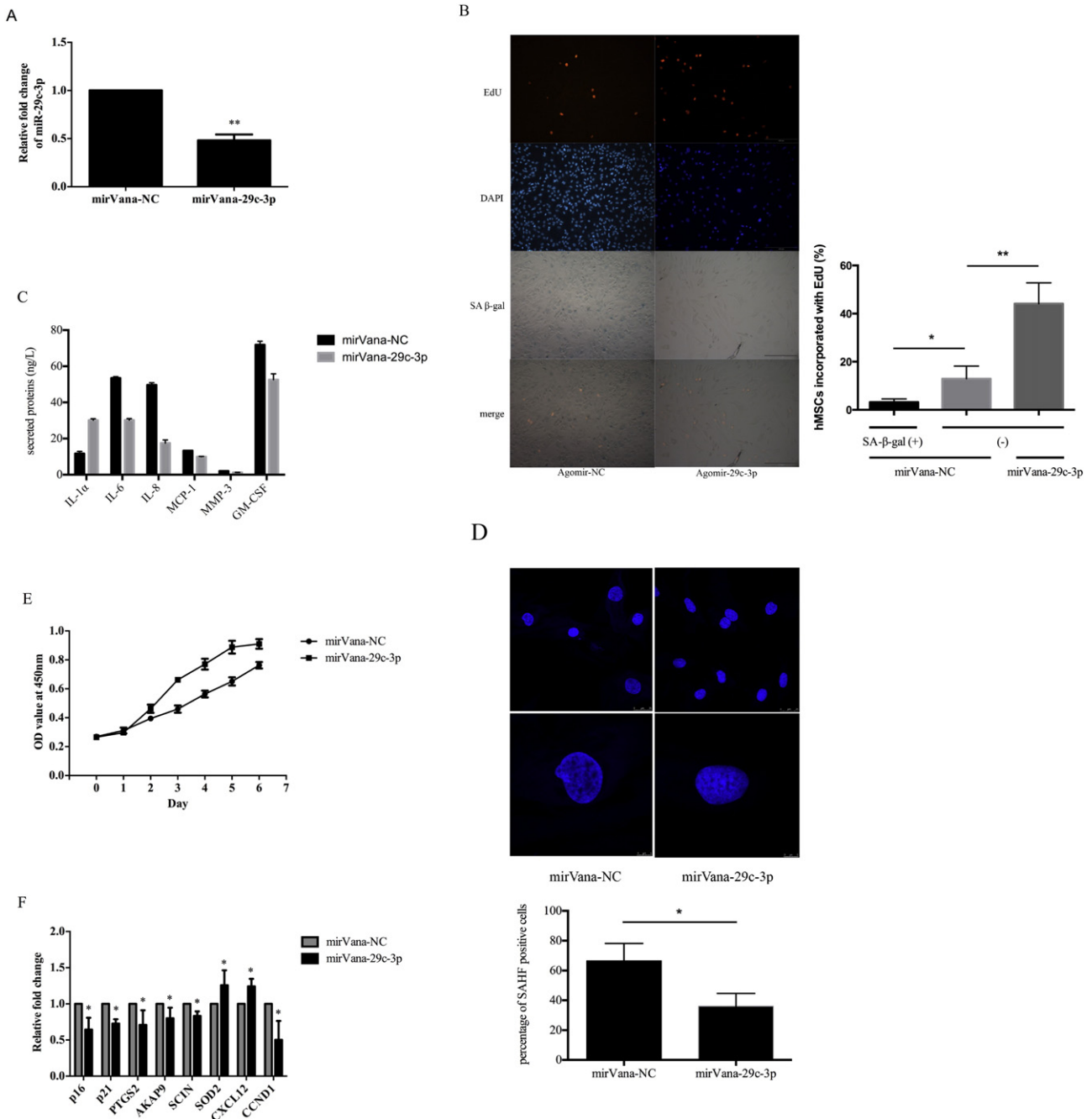


Fig. 3. Effects of miR-29c-3p inhibition on the senescence process of hMSCs. mirVana-29c-3p was transfected into hMSCs to down-regulate the expression of miR-29c-3p. mirVana-NC was used as a negative control. (A) After transfection, the expression of miR-29c-3p in the mirVana-NC and mirVana-29c-3p samples was compared using qPCR assays. (B) Co-staining of EdU and SA- β -gal was performed in mirVana-NC and mirVana-29c-3p samples. The histogram represented the percentages of EdU-positive cells between the two samples. (C) The SASP was detected by ELISA assays between mirVana-NC and mirVana-29c-3p groups. (D) The SAHF were visualized using DAPI staining between mirVana-NC and mirVana-29c-3p groups. The percentages of SAHF-positive hMSCs were calculated by counting the positive cells. The data were expressed as the mean $\pm$ S.D. (E) Changes in proliferative capacity due to miR-29c-3p inhibition were detected by CCK-8 assays. OD values were measured in triplicate at least every 24 h. A growth curve was obtained to compare the proliferative capacities between the samples. The expression levels of senescence-related genes after miR-29c-3p down-regulation were compared by (F) qPCR, (G) Western blot and (H) immunofluorescence (IF) assays. (I) Osteogenic differentiation was compared between mirVana-NC and mirVana-29c-3p samples using alizarin red staining. (J) Changes in the expression levels of osteogenic markers after miR-29c-3p inhibition were assessed by qPCR assays. Data were expressed as the mean $\pm$ standard deviation (S.D.). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

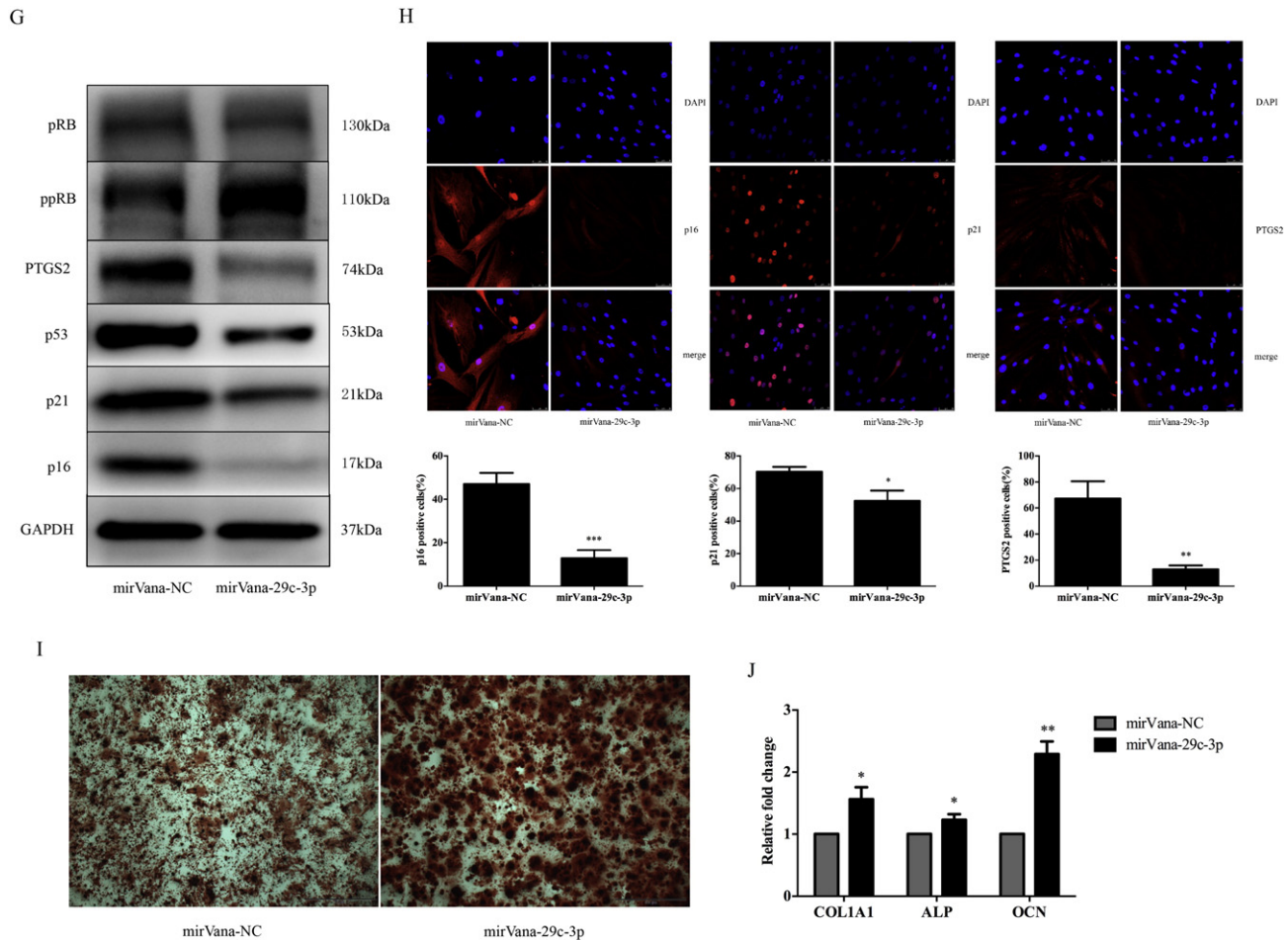


Fig. 3 (continued).

and stromal cells [24]. Fortunately, although there are no specific markers for MSC isolation, the nonclonal stromal cultures from bone marrow and other sources can still be treated as putative MSCs in preclinical studies and clinical trials [22]. However, the replicative senescence of hMSCs during *in vitro* expansion retards their physiological functions and therapeutic application [25]. The molecular mechanisms underlying hMSC senescence are not fully understood. miRNAs are a class of small and non-coding RNAs that are 18–25 nucleotides in length and generally degrade target gene expression at the post-transcriptional level [26,27]. Many studies have reported that miRNAs are involved in regulating cellular senescence [11,12,21,28]. miRNA-141-3p promotes the senescence of hMSCs by targeting ZMPSTE24 and results in prelamin A accumulation [11]. Moreover, miR-335 is found to be increased during hMSC senescence and suppresses their therapeutic actions by inhibiting AP-1 activity [12]. In addition, miR-17-3p acts as a negative regulator of mouse cardiac fibroblast senescence by targeting Par4 and the Par4-CEBPB-FAK senescence signaling pathway [13]. A microarray analysis revealed five significantly increased miRNAs in particular (hsa-miR-371, hsa-miR-369-3p, hsa-miR-29c, hsa-miR-499 and hsa-miR-217) during the replicative senescence of MSCs [14]. Therefore, we used qPCR assays to determine the miRNA expression levels in sequentially passaged hMSCs and found that miR-29c-3p was gradually up-regulated during extended periods of *in vitro* culture. This result logically suggested the genuine role of miR-29c-3p in promoting the replicative senescence of hMSCs. miR-29c has been reported to be involved in regulating cellular senescence. Carlos *et al.* found that the expression of miR-29 family members was strongly increased in *Zmpste24*^{-/-} progeroid mice as well as during

normal mouse aging. Moreover, miR-29, Ppm1d and p53 formed a regulatory circuit that regulated aging and the cellular response to DNA damage [15]. Additionally, miR-29 and miR-30 were found to be up-regulated during induced and replicative senescence, and they played an important role in cellular senescence by targeting B-Myb expression [16].

Here, we both increased and repressed miR-29c-3p expression in hMSCs to examine the following manifold cellular changes associated with replicative senescence, such as SA-β-gal activity, cell proliferation, and the expression of genes associated with replicative senescence and the osteogenic differentiation of hMSCs. Senescence is the process of aging at the cellular level [29]. Generally, SA-β-gal activity specifically increases in senescent cells. The deficiency of physiological functions usually results in suppressed cell proliferation and differentiation capacity. Generally, reports suggest that senescent cells display SASP, which changes cellular microenvironment and lead to age-related pathology by secreting growth factors and proteases [30]. SASP is considered as a cellular response to damage and differentiated among cell types and senescence stimulation [31,32]. We performed ELISA assays to detect miR-29c-3p related SASP. The results showed increased expression of IL-6, IL-8, MMP-3, MCP-1 and GM-CSF, except IL-1α, indicating inhibitory effect of IL-1α in miR-29c-3p induced senescence. Except for senescence-related secreted proteins, the changes of cell nuclei are also senescence markers. Cellular senescence is usually associated with spatial chromatin rearrangement and formation of nuclear structures, namely senescence-associated heterochromatic foci (SAHF). SAHF were once considered as a gene-silencing compartment [33,34]. However, SAHF don't emerge in all cellular senescence. The percentage

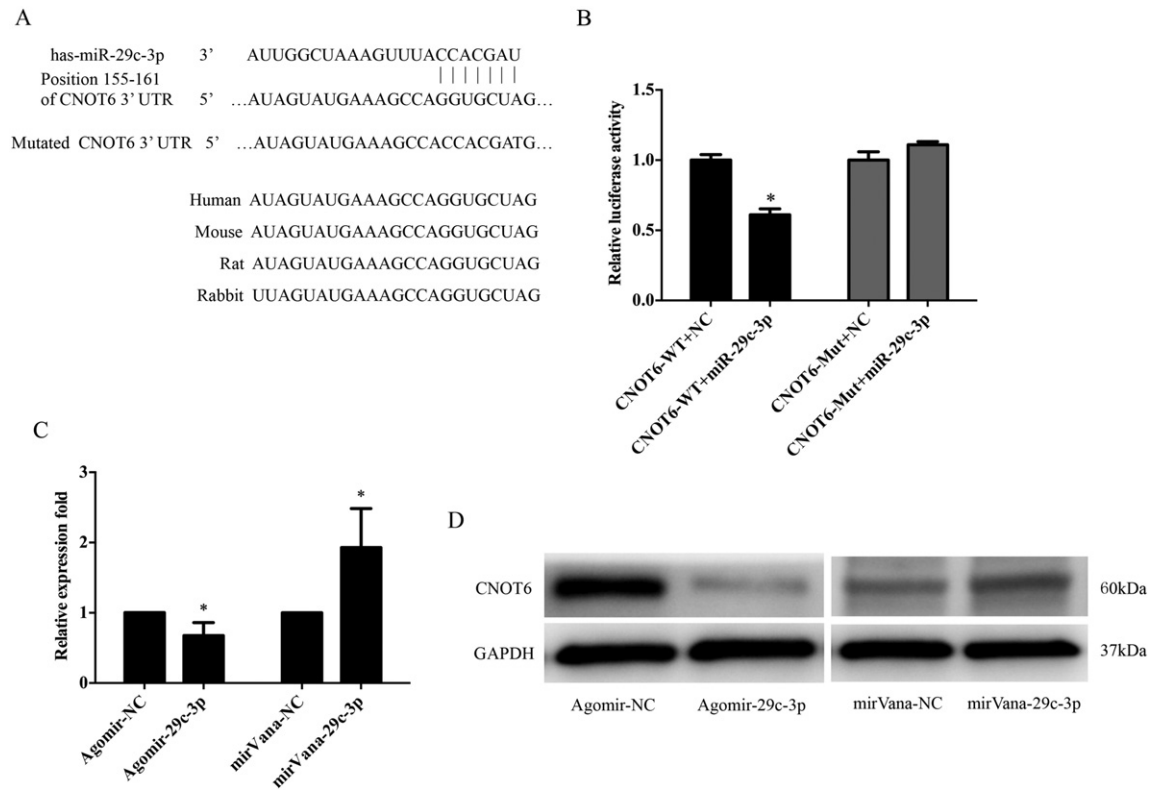


Fig. 4. CNOT6 was identified as the target gene of miR-29c-3p. (A) Complimentary base pairing between miR-29c-3p and the 3'UTR of humans and other mammals. (B) The results of dual-luciferase reporter assays were shown. The wild-type luciferase construct (CNOT6-WT) and the mutant luciferase construct (CNOT6-Mut) were each co-transfected with miR-29c-3p mimics and negative controls into 293 T cells. The luciferase activity was measured after a 48-h incubation. (C, D) The expression levels of CNOT6 after miR-29c-3p up-regulation and down-regulation were determined by qPCR and Western blot (WB) assays. The data were expressed as the mean $\pm$ standard deviation (S.D.). * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

of cells showing SAHF depend on the senescence stimulation method [35]. In our study, we found increased percentage of SAHF-positive cells in miR-29c-3p transfected hMSCs, suggesting the stimulative effect of miR-29c-3p on senescence. In addition, many genes have been considered as senescence markers or have been associated with replicative senescence, such as p16, p21, PTGS2, AKAP9, CCND1, SCIN, EDN1, CXCL12 and SOD2 [19,20,36,37]. p21 and p16 are both cyclin-dependent-kinase inhibitors, and they play different roles in the initiation and maintenance of senescence cell-cycle arrest [36]. Moreover, increases in PTGS2 (COX2) have been shown in both stress-induced and replicative senescence in fibroblasts, and it promotes the senescent phenotype [37–39]. Specifically, the expression of SOD2 is inversely correlated with cellular senescence. The reduction in SOD2 could impair mitochondrial activity and induce DNA damage, resulting in the promotion of senescence [20]. The results obtained in our study demonstrate that miR-29c-3p aggravates the replicative senescence of hMSCs, and its deficiency could rescue the senescence process.

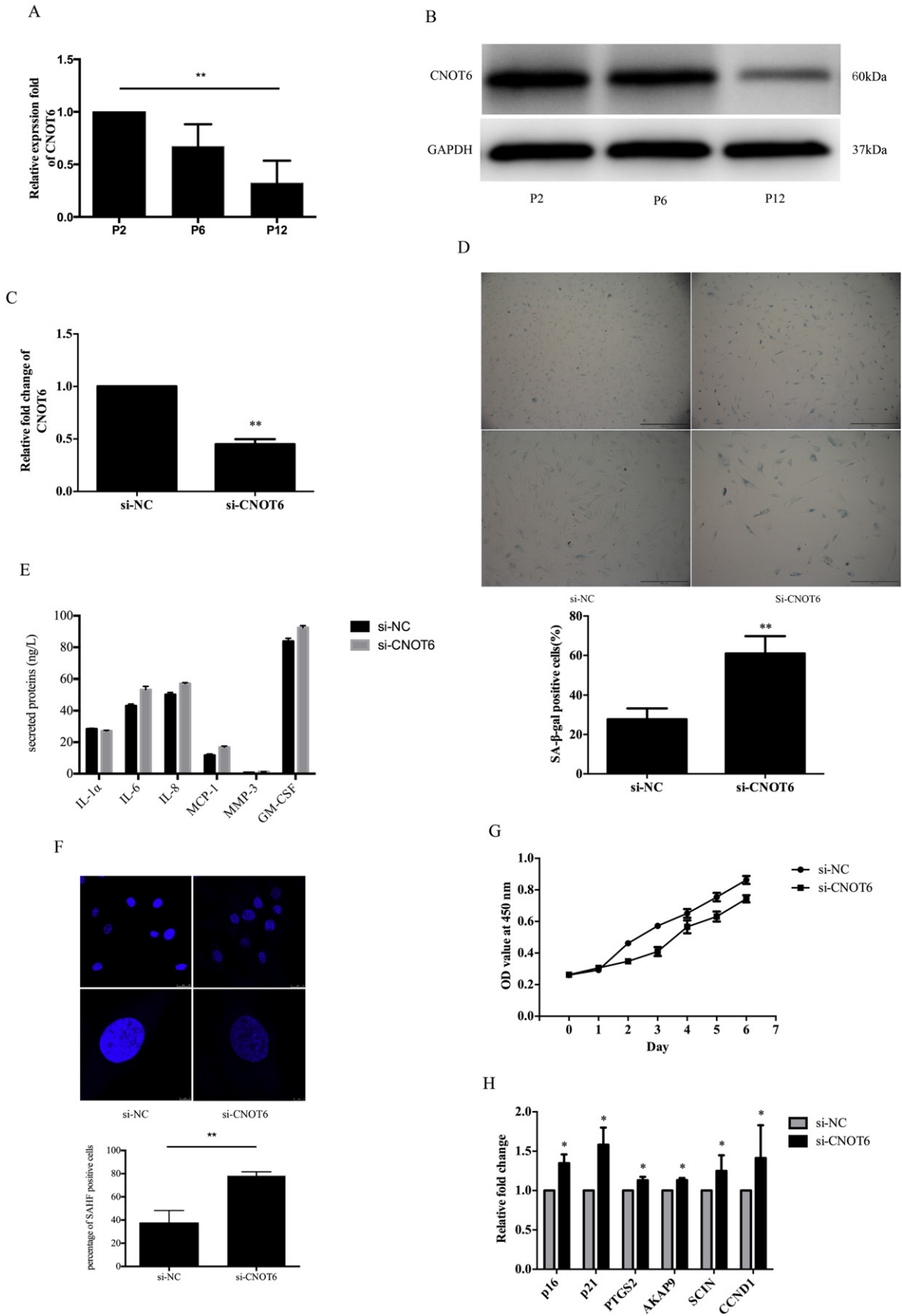
We found that the expression of CNOT6 gradually decreased as the hMSC passage number increased. Through bioinformatics analysis, we identified CNOT6 as the putative target gene of miR-29c-3p. The results of dual-luciferase reporter assays showed that miR-29c-3p directly bound to the 3'UTR of CNOT6 mRNA in hMSCs. CNOT6 is subunit 6 of the CCR4–NOT transcription complex. The CCR4–NOT transcription complex is one of the major cellular mRNA deadenylases and is involved in several cellular and physiological process, such as bulk mRNA degradation, miRNA-mediated repression and translational repression [40–42]. CNOT6 was found to prevent cell death and senescence [43]. CNOT6 and CNOT6L act as key regulators of insulin-like growth factor-binding protein 5 and prevent cellular senescence. In our

study, we used siRNA technology to knock down the endogenous expression of CNOT6. The results confirmed the promoted replicative senescence after CNOT6 down-regulation, suggesting the repressive effects of CNOT6 on cellular senescence.

Although the molecular mechanisms underlying cellular senescence are not clearly elucidated, generally, the emerging consensus believes that the p53–p21 pathway and p16–pRB pathway are the two major regulators of senescence [29,44–48]. The p53–p21 pathway mediates the replicative senescence and plays important roles in DNA damage response [49]. Alternatively, the p16–pRB pathway always mediates stress-induced and premature senescence [29,44,47]. However, certain cells cultured under specific conditions usually suffer from various endogenous and exogenous stimuli, so the activation of both major pathways in senescence may differ depending on the specific circumstance. In our study, both the p53–p21 pathway and the p16–pRB pathway were activated in sequentially passaged hMSCs. We next increased miR-29c-3p expression by transfecting Agomir-29c-3p into hMSCs, and found that both the p53–p21 and the p16–pRB pathways were activated. Contrarily, after down-regulation of miR-29c-3p in hMSCs, both the major pathways were suppressed. The results indicated that miR-29c-3p exerted its effects on hMSCs senescence through the two master pathways. Typically, p53 and pRB are activated in senescence, and either of them could induce cellular senescence in certain cell types when increasing their expression levels using genetic methods [50,51]. The block of these two pathways could abrogate cellular senescence of human and mouse cells, despite the initiation of senescence [29,44,52]. The p21 protein, one of the cell cycle inhibitors, is the downstream target gene of the p53–p21 pathway. In addition, the p16–pRB pathway generally consists of four proteins–p16INK4a, cyclin D1, cdk4, and pRB [53]. The p16INK4a inhibits cyclinD1/cdk4 kinase to

activate pRB. Generally, the pRB pathway retard cell growth though several downstream effectors. For instance, pRB suppresses E2F transcription factor family, of which target genes are essential for

progression into S phase [53]. In senescent cells, when the mitogenic signal is blocked, pRB becomes unphosphorylated and suspends the cell cycle, leaving the cells in G₁ [54,55]. The retinoblastoma family



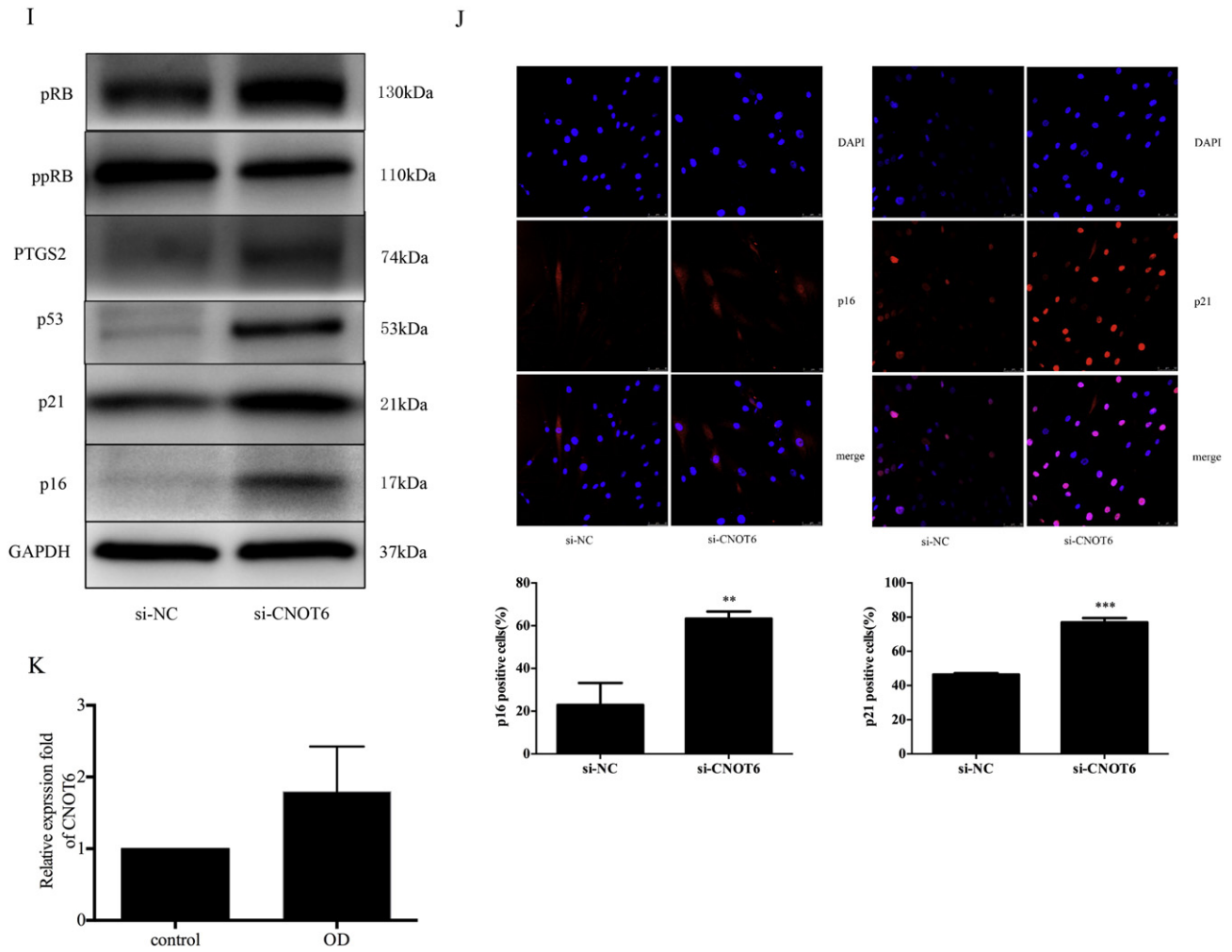


Fig. 5. Effects of CNOT6 down-regulation on the senescence of hMSCs. The expression of CNOT6 in sequentially passed hMSCs (P2, P6 and P12) was measured by qPCR (A) and Western blot (WB) (B) assays. si-CNOT6 was transfected into hMSCs to inhibit the expression of endogenous CNOT6. si-NC was used as a negative control. (C) After transfection, the expression levels of CNOT6 were compared between si-NC and si-CNOT6 samples using qPCR assays. (D) The SA- β -gal activity was compared between si-NC and si-CNOT6 samples. The histogram compared the rates of SA- β -gal-positive cells between the two samples. (E) The SASP was detected by ELISA assays between si-NC and si-CNOT6 groups. (F) The SAHF were visualized using DAPI staining between si-NC and si-CNOT6 groups. The percentages of SAHF-positive hMSCs were calculated by counting the positive cells. The data were expressed as the mean $\pm$ S.D. (G) Changes in the proliferation capacity due to CNOT6 inhibition were detected by CCK-8 assays. The OD values were measured in triplicate at least every 24 h. A growth curve was obtained to compare the proliferative capacities between the samples. The expression levels of senescence-related genes after miR-29c-3p down-regulation were compared by (H) qPCR, (I) Western blot and (J) immunofluorescence (IF) assays. The data were expressed as the mean $\pm$ S.D. (K) The change of CNOT6 expression during osteogenic differentiation was investigated by qPCR assays. The data were expressed as the mean $\pm$ S.D. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

genes include RB1/pRB, RB2/p130 and p107 and play vital roles in controlling the cell cycle G₁/S transition. However, the functions of retinoblastoma genes are found to be controversial. RB1/pRB is considered to control the G₁/S checkpoint and involved in senescent arrest [33]. In malignant tumors, decrease of RB1/pRB was thought to abolish the senescence barrier and possibly enhance malignant transformation [56]. But Shamma *et al.* found that the depletion of RB1/pRB induced cellular senescence *in vitro* and *in vivo* [57]. Additionally, only RB2/p130 increased in senescent cells and suppressed the promoter of S-phase genes [58]. Further studies need to address the molecular mechanisms of senescence in details.

In summary, our study demonstrates that miR-29c-3p is involved in promoting the senescence of hMSCs by targeting CNOT6. Moreover, both the p53-p21 and p16-pRB signaling pathways are activated in miR-29c-3p-induced senescence. Understanding senescence-associated miRNAs and genes is beneficial for elucidating the mechanisms of cellular senescence and enhancing applications of MSCs in tissue engineering and clinical therapies.

Conflict of interest

The authors declare that they have no conflict of interest.

Ethical standards

The experiments comply with the current laws of China in which they were performed.

Transparency document

The Transparency document associated with this article can be found, in the online version.

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